6.3 Pascal's Triangle – Worksheet #1

MCR3U

SOLUTIONS.

1) Find each coefficient described.

a) coefficient of x^2 in the expansion of $(2+x)^5$

3rd term is $10(2)^3(x)^2 = 80x^2$

∴ the coefficient is 80.

b) coefficient of x^2 in the expansion of $(x+2)^5$

4th term is $10(x)^2(2)^3 = 80x^2$

∴ the coefficient is 80.

c) coefficient of x in the expansion of $(x+3)^5$

5th term is $5(x)^1(3)^4 = 405x$

∴ the coefficient is 405

d) coefficient of b in the expansion of $(3+b)^4$

2nd term is $4(3)^3(b) = 108b$

∴ the coefficient is 108.

e) coefficient of x^3y^2 in expansion of $(x-3y)^5$

3rd term is $10(x)^3(-3y)^2 = 90x^3y^2$

∴ the coefficient is 90

f) coefficient of a^2 in the expansion of $(2a+1)^5$

4th term is $10(2a)^2(1)^3 = 40a^2$

∴ the coefficient is 40.

2) Find each term described.

a) 2nd term in expansion of $(y-2x)^4$

$= 4(y)^3(-2x)^1$

$= -8y^3x$

b) 4th term in expansion of $(4y+x)^4$

$= 4(4y)(x)^3$

$= 16yx^3$

c) 1st term in expansion of $(a+b)^5$

$= 1(a)^5(b)^0$

$= a^5$

d) 2nd term in expansion of $(y-x)^4$

$= 4(y)^3(-x)^1$

$= -4y^3x$

3) Expand completely

a) $(2m-1)^4$

$= 1(2m)^4 + 4(2m)^3(-1) + 6(2m)^2(-1)^2 + 4(2m)(-1)^3 + 1(-1)^4$

$= 16m^4 - 32m^3 + 24m^2 - 8m + 1$

b) $(x-y)^3$

$= 1(x)^3 + 3(x)^2(-y) + 3(x)(-y)^2 + 1(-y)^3$

$= x^3 - 3x^2y + 3xy^2 - y^3$

c) $(x^4 - y)^5$

$= 1(x^4)^5 + 5(x^4)^4(-y) + 10(x^4)^3(-y)^2$
$+ 10(x^4)^2(-y)^3 + 5(x^4)(-y)^4 + 1(-y)^5$

$= x^{20} - 5x^{16}y + 10x^{12}y^2 - 10x^8y^3$
$+ 5x^4y^4 - y^5$

d) $(2x^3 + 1)^5$

$= 1(2x^3)^5 + 5(2x^3)^4(1) + 10(2x^3)^3(1)^2$
$+ 10(2x^3)^2(1)^3 + 5(2x^3)(1)^4 + 1(1)^5$

$= 32x^{15} + 80x^{12} + 80x^9 + 40x^6 + 10x^3 + 1$

e) $(y - x^2)^3$

$= 1(y)^3 + 3(y)^2(-x^2)^1 + 3(y)^1(-x^2)^2 + 1(-x^2)^3$

$= y^3 - 3y^2x^2 + 3yx^4 - x^6$

f) $(y^3 - 4x)^3$

$= 1(y^3)^3 + 3(y^3)^2(-4x)^1 + 3(y^3)^1(-4x)^2 + 1(-4x)^3$
$= y^9 - 12y^6x + 48y^3x^2 - 64x^3$

Answers

1) a) 80 **b)** 80 **c)** 405 **d)** 108 **e)** 90 **f)** 40

2) a) $-8y^3x$ **b)** $16yx^3$ **c)** a^5 **d)** $-4y^3x$

3) a) $16m^4 - 32m^3 + 24m^2 - 8m + 1$ **b)** $x^3 - 3x^2y + 3xy^2 - y^3$

c) $x^{20} - 5x^{16}y + 10x^{12}y^2 - 10x^8y^3 + 5x^4y^4 - y^5$

d) $32x^{15} + 80x^{12} + 80x^9 + 40x^6 + 10x^3 + 1$

e) $y^3 - 3y^2x^2 + 3yx^4 - x^6$ **f)** $y^9 - 12y^6x + 48y^3x^2 - 64x^3$